The Model Equations

Part 1 - Logistic regression (threshold decision)

- Models whether cong_t crosses zero.
- Whether ESA listing crosses the congressional attention threshold this month.
- Uses all 371 months.
- The logistic regression produces one probability for every month in the dataset.

P(cong_t > 0) = exp(eta) / (1 + exp(eta))

eta = gamma_0 + gamma_4 x news_lag + gamma_3 x cong_lag

Part 1: Threshold Probabilities for all Observed news_lag / cong_lag combinations

news_lag	cong_lag	eta	P(active)	P(silent)	n months
0	0	−3.651	2.5%	97.5%	333
0	1	−3.136	4.2%	95.8%	8
0	2	−2.620	6.8%	93.2%	1
0	5	−1.074	25.5%	74.5%	1
1	0	−2.611	6.8%	93.2%	23
2	0	−1.571	17.2%	82.8%	2
2	1	−1.056	25.8%	74.2%	1
3	0	−0.531	37.0%	63.0%	2

Logistic coefficients

Parameter	Estimate	Std. Error	exp(Est.)
gamma_0 (Intercept)	-3.6510	0.3349	0.026
gamma_4 news_lag	+1.0400	0.3996	2.829
gamma_3 cong_lag	+0.5153	0.4352	1.674

Primary finding — gamma_4 = +1.040: exp(1.040) = 2.83. Each additional ESA listing news article last month multiplies the odds of any congressional activity by 2.83. This directly answers the agenda-setting hypothesis at the threshold level. **P-value < 0.001 — highly significant.**

Part 2 — Zero-truncated Poisson (count intensity)

- Models how many documents Congress produces given cong_t > 0.
- Uses only the 12 active months.
- The 359 zero months do not enter Part 2 at all.

cong_t | cong_t > 0 ~ Poisson(mu)

log(mu) = beta_0 + beta_4 x news_lag + beta_3 x cong_lag

The active 12 months where cong_t > 0

Month	cong_t	news_lag	cong_lag
1995-05	**1**	**0**	**0**
1998-04	**1**	**0**	**0**
1999-04	**1**	**3**	**0**
1999-06	**1**	**0**	**0**
1999-07	**1**	**2**	**1**
2015-07	**2**	**0**	**0**
2016-07	**1**	**0**	**0**
2024-03	**1**	**0**	**0**
2025-01	**5**	**0**	**0**
2025-03	**1**	**0**	**0**
2025-04	**1**	**0**	**1**
2025-12	**1**	**0**	**0**

Poisson coefficients

Question answered: Given that Congress is active, does media coverage predict **how many** documents it produces?

Parameter	Estimate	Std. Error	exp(Est.)
beta_0 (Intercept)	-0.041	0.420	0.960
beta_4 news_lag	-5.362	1497.9	0.005
beta_3 cong_lag	-13.787	1422.6	~0.000

Findings:

Verification for April 1999

April 1999 had news_lag = 3, cong_lag = 0, and actual cong_t = 1.

Step 1 — Part 1 (logistic):

eta = -3.6510 + 1.0400 x 3 + 0.5153 x 0 = -0.5311

P(cong_t > 0) = exp(-0.5311) / (1 + exp(-0.5311)) = 0.3703

Interpretation: Part 1 gives a 37.0% probability that Congress crosses the attention threshold in April 1999.

Step 2 — Part 2 (Poisson, given active):

log(mu) = -0.041 + (-5.362) x 3 + (-13.787) x 0 = -16.127

mu = exp(-16.127) ≈ 0.000

Plugging in the actual numbers from April 1999

mu = exp(-16.127) = 0.0000000986

P(cong_t = 1 | active) = [exp(-mu) x mu] / [1 - exp(-mu)]

= [0.9999999 x 0.0000000986] / 0.0000000986

~ 0.9999 (99.99%)

The 99.99% figure means that, conditional on Congress already being active in April 1999, the zero-truncated Poisson distribution in Part 2 assigns virtually all of its probability mass to the count being exactly 1.

What April 1999 shows about the Two-Question Structure

Question	Coefficient	Result
gamma_4 — does media predict WHETHER Congress engages?	gamma_4 = 1.040	P(active) rises from 2.5% to 37.0%
beta_4 — does media predict how much Congress produces, given it engages?	beta_4 = -5.362	mu collapses toward 0 given high news_lag

REFERENCES

https://library.virginia.edu/data/articles/getting-started-with-hurdle-models

https://search.r-project.org/CRAN/refmans/pscl/html/hurdle.html

Regression Models for Count Data in R